PROBABILITY FOR COMPUTING

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UNIVERSITY OF DELHI

PRACTICAL FILE

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SEMESTER – 2

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INDEX

S. No.	Topic	Page No.
i	Title Page	
ii	Acknowledgement	
iii	Index	
1.	Plotting and fitting of Binomial distribution and graphical representation of probabilities.	
2.	Plotting and fitting of Multinomial distribution and graphical representation of probabilities.	
3.	Plotting and fitting of Poisson distribution and graphical representation of probabilities.	
4.	Plotting and fitting of Geometric distribution and graphical representation of probabilities.	
5.	Plotting and fitting of Uniform distribution and graphical representation of probabilities.	
6.	Plotting and fitting of Exponential distribution and graphical representation of probabilities.	
7.	Plotting and fitting of Normal distribution and graphical representation of probabilities.	
8.	Calculation of cumulative distribution functions for Exponential and Normal distribution.	
9.	Given data from two distributions, find the distance between the distributions.	
10.	Application problems based on the Binomial distribution.	
11.	Application problems based on the Poisson distribution.	
12.	Application problems based on the Normal distribution.	

S. No.	Topic	Page No.
13.	Presentation of bivariate data through scatter-plot diagrams and calculations of covariance.	
14.	Calculation of Karl Pearson's correlation coefficients.	
15.	To find the correlation coefficient for a bivariate frequency distribution.	
16.	Generating Random numbers from discrete (Bernoulli, Binomial, Poisson) distributions.	
17.	Generating Random numbers from continuous (Uniform, Normal) distributions.	
18.	Find the entropy from the given data set.	

Practical 1: Plotting and Fitting of Binomial Distribution and Graphical Representation of Probabilities

Theory

The binomial distribution is one of the most fundamental discrete probability distributions in statistics. It is used to determine the likelihood of a specific number of successes occurring across a fixed number of independent trials, where every trial can yield one of only two possible outcomes — a success or a failure. This distribution operates under the assumption that each trial is statistically independent and that the probability of achieving a success remains unchanged throughout all trials.

The binomial distribution is characterized by two key parameters:

- **n**: represents the total number of trials conducted
- **p**: denotes the probability of achieving a success in any individual trial

Binomial Distribution Formula

```
P(X = k) = C(n, k) * p^k * (1 - p)^(n-k)
```

- **Mean:** mu = E(X) = n x p
- **Variance:** sigma^2 = V(X) = n x p x (1 - p)

Where:

- n = total number of independent trials
- k = number of successes observed
- p = probability of success on any given trial
- C(n, k) = n! / [k! x (n - k)!] (the binomial coefficient)

How to Implement in Excel

Excel Formula: `=BINOM.DIST(k, n, p, FALSE)`

Dataset

The following tables display the binomial distribution probabilities for **n = 10** trials under three different values of the success probability p.

Case 1: n = 10, p = 0.7 (Left-skewed; bulk of probability concentrated at higher success counts)

k (Successes)	Binomial Probability
0	0.00000590
1	0.00013778
2	0.00144670
3	0.00900169

4	0.03675691
5	0.10291935
6	0.20012095
7	0.26682793
8	0.23347444
9	0.12106082
10	0.02824752

Excel formula (Case 1): `=BINOM.DIST(k, 10, 0.7, FALSE)`

When p = 0.7, the distribution is skewed to the left. The bulk of probability mass is concentrated in the higher success counts (k = 6-8), with the distribution tailing off toward the lower end.

Case 2: n = 10, p = 0.5 (Symmetric distribution)

k (Successes)	**Binomial Probability**
0	0.00097656
1	0.00976563
2	0.04394531
3	0.11718750
4	0.20507813
5	0.24609375
6	0.20507813
7	0.11718750
8	0.04394531
9	0.00976563
10	0.00097656

Excel formula (Case 2): `=BINOM.DIST(k, 10, 0.5, FALSE)`

When p = 0.5, the distribution exhibits perfect symmetry around the central value, n/2 = 5.

Case 3: n = 10, p = 0.3 (Right-skewed; concentrated at lower success counts)

k (Successes)	**Binomial Probability**
0	0.02824752
1	0.12106082
2	0.23347444
3	0.26682793
4	0.20012095
5	0.10291935
6	0.03675691
7	0.00900169
8	0.00144670
9	0.00013778
10	0.00000590

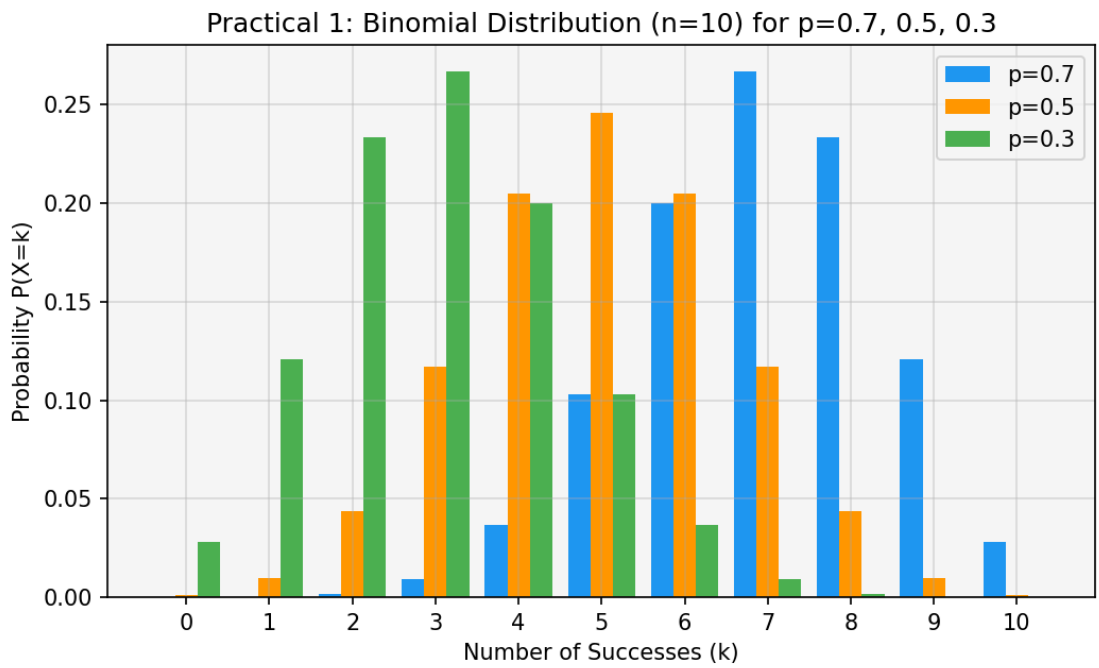

Figure 1: Binomial Distribution (n=10) — P(X=k) for p=0.7, 0.5 and 0.3

Excel formula (Case 3): `=BINOM.DIST(k, 10, 0.3, FALSE)`

When p = 0.3, the distribution is right-skewed, with the highest probabilities occurring at small values of k (k = 2-3).

Practical 2: Plotting and Fitting of Multinomial Distribution and Graphical Representation of Probabilities

Theory

The multinomial distribution is a multivariate extension of the binomial distribution. It applies when an experiment can result in exactly one of k possible distinct outcomes per trial, with each outcome i having a fixed probability p_i (where the sum of all probabilities equals 1), and there are n statistically independent trials. The random variables X_1, X_2, ..., X_k each record how many times the outcome was observed across the n trials.

When the joint vector X = (X_1, X_2, ..., X_k) follows a multinomial distribution with parameters n and p = (p_1, p_2, ..., p_k), each individual X_i follows a marginal binomial distribution with n trials and success probability p_i.

Multinomial Distribution Formula

```
P(x1, x2, ..., xk) = [n! / (x1! * x2! * ... * xk!)] * p1^x1 * p2^x2 * ...
* pk^xk
```

Covariance: Cov(Xi, Xj) = -n * pi * pj (for i != j)

How to Implement in Excel

- **Multinomial coefficient:** `=MULTINOMIAL(X1, X2, X3)`
- **Probability:** `=MULTINOMIAL(X1,X2,X3) * PRODUCT(p1^X1, p2^X2, p3^X3)`

Dataset

The table below presents three different scenarios involving Red, Green, and Blue balls drawn from a container.

Parameters are held constant: **p1 = 0.40**, **p2 = 0.25**, **p3 = 0.35**, **n = 10**.

Parameter	**Case 1**	**Case 2**	**Case 3**
Red (X1)	4	3	5
Green (X2)	3	2	2
Blue (X3)	3	5	3
P_Red (P1)	0.40	0.40	0.40
P_Green (P2)	0.25	0.25	0.25
P_Blue (P3)	0.35	0.35	0.35
Multinomial Coefficient	4200	2520	2520

Parameter	**Case 1**	**Case 2**	**Case 3**
P1^X1	0.025600	0.064000	0.010240
P2^X2	0.015625	0.062500	0.062500
P3^X3	0.042875	0.005252	0.042875
Probability	**0.072030**	**0.052942**	**0.069149**

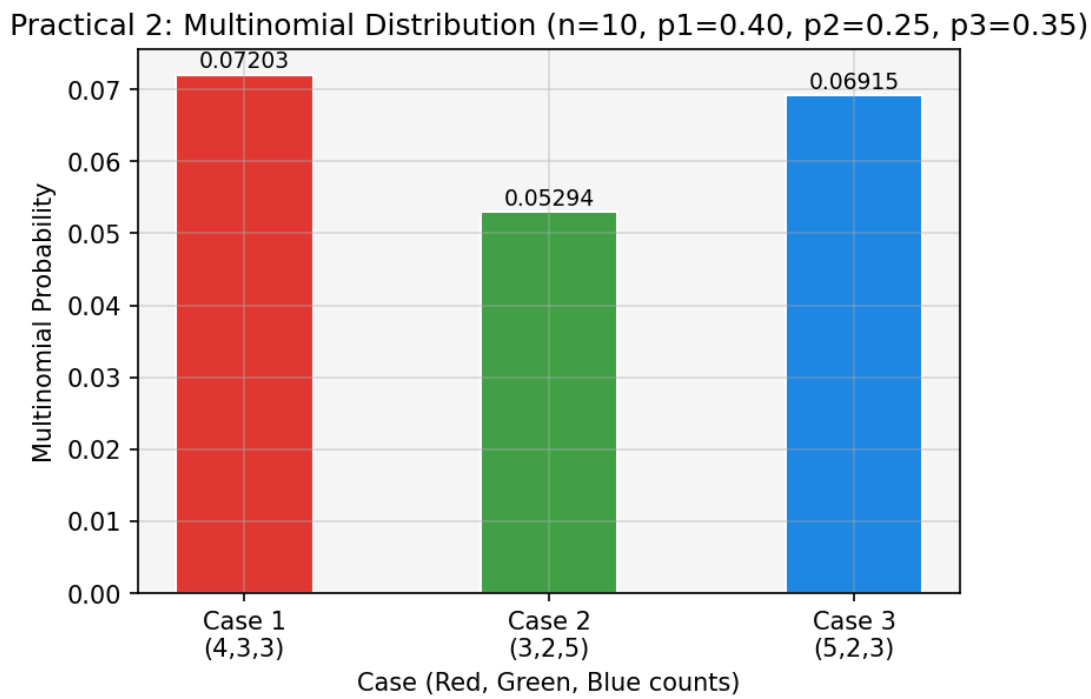

Figure 2: Multinomial Distribution — Probability by Case (n=10)

Excel formula (Case 1): `=MULTINOMIAL(4,3,3) * PRODUCT(0.4^4, 0.25^3, 0.35^3)`

Note: The probabilities in the three cases illustrate how varying the count of each outcome type affects the multinomial probability while holding the outcome probabilities constant.

Practical 3: Plotting and Fitting of Poisson Distribution and Graphical Representation of Probabilities

Theory

The Poisson distribution is a discrete probability distribution that quantifies the probability of a given number of events occurring within a fixed time interval or spatial region. The distribution assumes that events occur continuously and independently at a constant

average rate. It is governed by a single parameter, **lambda**, which represents the average number of event occurrences within the specified interval.

Poisson Distribution Formula

```
P(k) = e^(-lambda) * lambda^k / k!
```

- **Mean:** mu = E(X) = lambda
- **Variance:** sigma^2 = V(X) = lambda
- **Standard Deviation:** sigma = sqrt(lambda)

Where:

- P(X = k) is the probability of observing exactly k events
- e is Euler's number (approximately 2.71828)
- lambda is the mean number of events occurring during a specific interval; lambda = np
- k is the number of events (successes) of interest

How to Implement in Excel

Excel Formula: `=POISSON.DIST(k, lambda, FALSE)`

Dataset

The table below shows the Poisson distribution for **lambda = 6** (mean number of successes), evaluated for k = 0 to 10:

k (Successes)	Poisson Distribution P(X=k)
0	0.002479
1	0.014873
2	0.044618
3	0.089235
4	0.133853
5	0.160623
6	0.160623
7	0.137677
8	0.103258
9	0.068838
10	0.041303

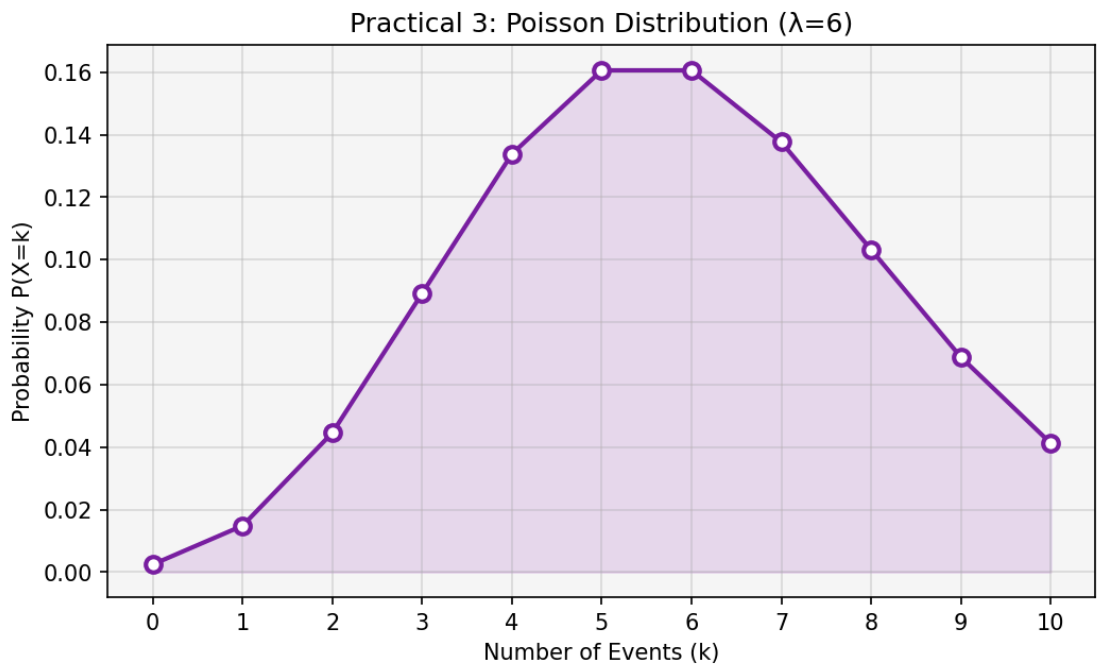

Figure 3: Poisson Distribution (λ=6) — P(X=k) for k=0 to 10

Excel formula: `=POISSON.DIST(k, 6, FALSE)` *for each k value.*

When lambda = 6, the distribution peaks at k = 5 and k = 6 (equal probabilities), after which the probabilities decrease steadily.

Practical 4: Plotting and Fitting of Geometric Distribution and Graphical Representation of Probabilities

Theory

In a Bernoulli trial, a geometric distribution models the number of consecutive failures that are observed before the first success is recorded. A Bernoulli trial is defined as a random experiment that can produce exactly one of two outcomes: success or failure. The distinguishing characteristic of a geometric distribution is that the process of repeated Bernoulli trials continues until a success is obtained, at which point it is terminated.

The geometric distribution is a discrete probability distribution that captures the likelihood of needing exactly k failures before the first success occurs. The number of attempts may

theoretically extend without bound until the initial success is achieved. This distribution rests on three fundamental assumptions:

- The individual trials being conducted are mutually independent.
- Each trial must result in exactly one of two possible outcomes: either a success or a failure.
- The probability of success, denoted by p, remains fixed and identical across every trial.

Geometric Distribution Formula

```
P(X = k) = (1 - p)^k * p
```

- **Mean:** mu = E(X) = (1 - p) / p
- **Variance:** sigma^2 = V(X) = (1 - p) / p^2

Where:

- k: the count of failures that occur prior to the first success
- p: the probability of achieving success on each individual trial

The probability of a successful outcome is represented by p, whereas the probability of failure on any given trial is denoted by q, where q = 1 - p. The notation X ~ G(p) signifies that the random variable X follows a geometric probability distribution.

How to Implement in Excel

Excel Formula: `= (1 - p)^k * p`

Dataset

The following table presents the geometric distribution for **p = 0.25** (probability of success on each trial):

Failures before 1st Success (k)	Geometric Probability P(X=k)
0	0.250000
1	0.187500
2	0.140625
3	0.105469
4	0.079102
5	0.059326
6	0.044495
7	0.033371
8	0.025028
9	0.018771
10	0.014078

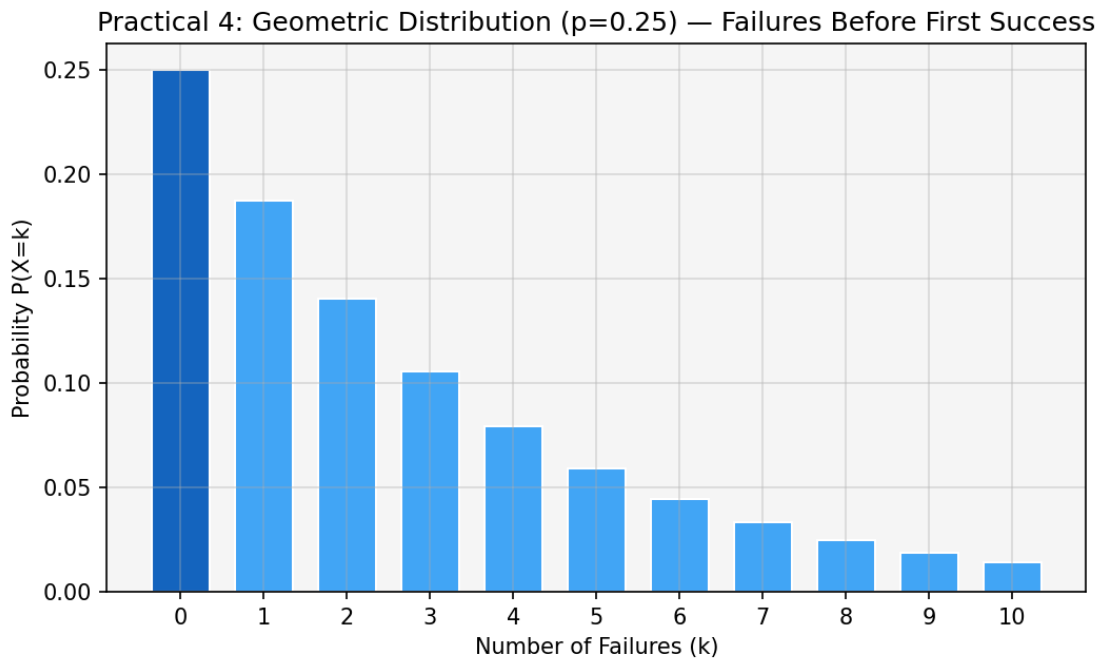

Figure 4: Geometric Distribution (p=0.25) — Failures Before First Success

Excel formula: `=(1-0.25)^k*0.25` *or equivalently* `=0.75^k*0.25` *for each k.*

With p = 0.25, the highest probability occurs at k = 0 (immediate first-try success). As k increases, the probability decreases steadily, reflecting the diminishing chance of extended failure streaks.

Practical 5: Plotting and Fitting of Uniform Distribution and Graphical Representation of Probabilities

Theory

A uniform distribution is a type of probability distribution in which every outcome within a specified range is equally likely to occur. It is also commonly referred to as a rectangular distribution (in the context of the continuous case). The distribution is fully characterized by two parameters: **a** (the minimum value) and **b** (the maximum value), and is expressed as U(a, b).

In a uniform distribution, every possible value within the defined range carries an identical probability of being observed. This property ensures that no value within the interval [a, b] is inherently more likely than any other.

Uniform Distribution Formula

The probability density function (PDF) of a continuous uniform distribution defines the probability that a random variable falls within a particular sub-interval of [a, b]:

```
f(x) = 1 / (b - a)     for a <= x <= b
Mean: mu = (a + b) / 2
Variance: sigma^2 = (b - a)^2 / 12
```

How to Implement in Excel

Excel Formula for Probability: `P = (x2 - x1) / (b - a)`

To compute the probability that a uniform random variable falls between x1 and x2, the following four inputs are required:

- **a**: the minimum value of the distribution
- **b**: the maximum value of the distribution
- **x1**: the lower bound of the interval of interest
- **x2**: the upper bound of the interval of interest

Dataset

The following table presents three cases using **a = 5** and **b = 25** (total range = 20):

Parameter	**Case 1**	**Case 2**	**Case 3**
a (Min)	5	5	5
b (Max)	25	25	25
x1	8	12	15
x2	14	20	22
Interval Width (x2 - x1)	6	8	7
Uniform Distribution P	**0.3000**	**0.4000**	**0.3500**

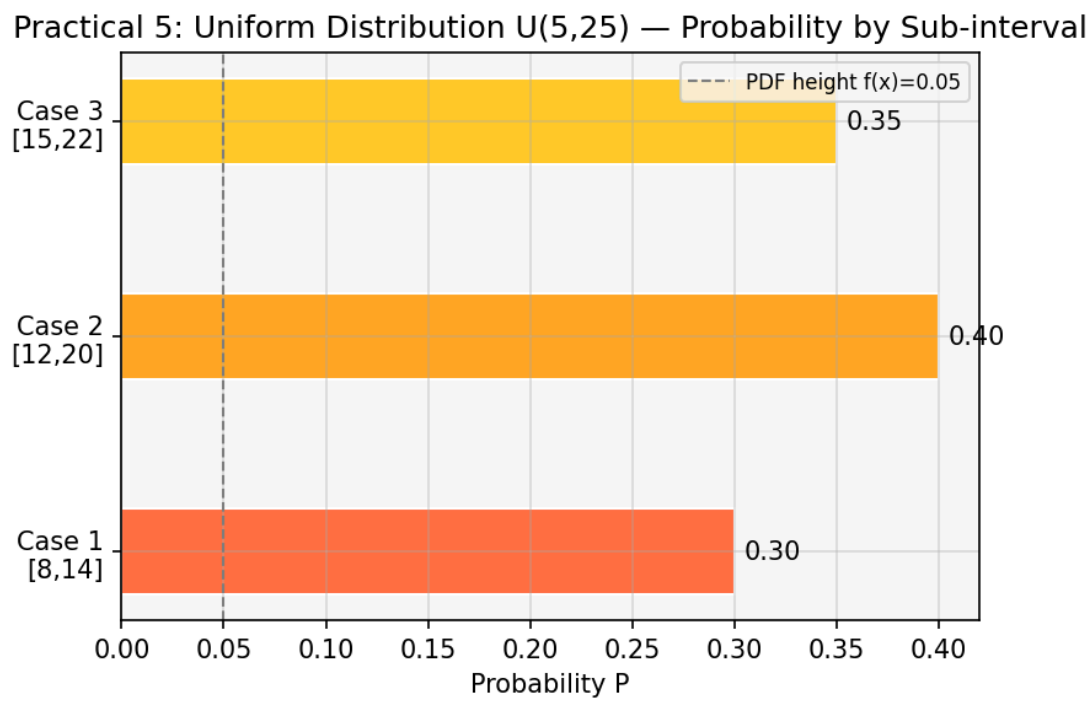

Figure 5: Uniform Distribution U(5,25) — Probability by Sub-interval

Excel formula (Case 1): `=(14-8)/(25-5)` *= 0.30*

As shown in the table, the probability is directly proportional to the width of the sub-interval (x2 - x1) relative to the total range (b - a = 20).

Practical 6: Plotting and Fitting of Exponential Distribution and Graphical Representation of Probabilities

Theory

The exponential distribution is a continuous probability distribution defined over the set of all non-negative real numbers. It is particularly suited for modeling the time elapsed before a given event occurs. When we ask questions such as 'How long must we wait until an event happens?', the answer can be modelled probabilistically using the Exponential Distribution. Since the waiting time is unknown in advance, it is treated as a continuous Random Variable. If the probability of the event occurring during any time interval is proportional to the length of that interval, the Random Variable follows an exponential distribution.

This distribution is commonly applied to the following types of real-world problems:

- How long does a shopkeeper need to wait before a customer enters the store.
- How long will a battery continue to function before it fails.
- How long will a computer system remain operational before it encounters a breakdown.

Exponential Distribution Formula

```
f(x) = lambda * e^(-lambda*x)     for x >= 0
f(x) = 0                          for x < 0

E(X) = 1 / lambda
Var(X) = 1 / lambda^2
```

Where lambda is the rate parameter, which controls the shape and spread of the density function. The constant e is approximately 2.718.

How to Implement in Excel

Excel Formula: `=EXPON.DIST(X, lambda, FALSE)`

Dataset

The table below presents the exponential PDF values for **mu = 333.33** and **lambda = 0.003** across three different x values:

Parameter	**Case 1**	**Case 2**	**Case 3**
mu (Mean = 1/lambda)	333.33	333.33	333.33
lambda	0.003	0.003	0.003
x	400	450	500
Exponential PDF f(x)	**0.000904**	**0.000778**	**0.000669**

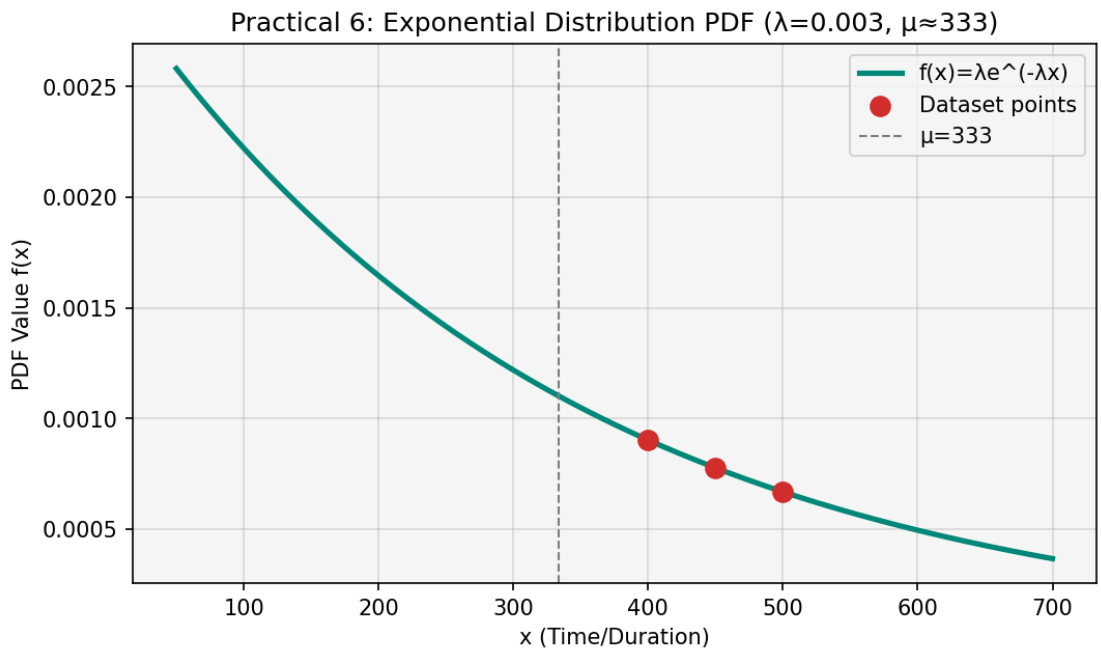

Figure 6: Exponential Distribution PDF (λ=0.003, μ≈333)

Excel formula (Case 1): `=EXPON.DIST(400, 0.003, FALSE)`

As x increases beyond the mean mu approx 333, the exponential PDF decreases progressively, consistent with the decaying nature of the exponential density function.

Practical 7: Plotting and Fitting of Normal Distribution and Graphical Representation of Probabilities

Theory

The Normal Distribution, also known as the Gaussian distribution, is defined as the probability density function of any continuous random variable within a given system. It describes the distribution of a random variable X whose behavior conforms to a characteristic bell-shaped curve. The Normal distribution is entirely defined by two parameters: the mean (mu) and the standard deviation (sigma).

Normal Distribution Formula

```
f(x) = [1 / (sigma * sqrt(2*pi))] * e^[-(1/2) * ((x - mu)/sigma)^2]
```

Where:

- x is the Random Variable
- mu is the Mean of the distribution
- sigma is the Standard Deviation

Properties of Normal Distribution

- For normally distributed data, the mean, median, and mode are all equal (Mean = Median = Mode).
- The total area enclosed beneath the normal distribution curve is always equal to 1.
- The normal distribution curve is symmetric about its central axis, which passes through the mean.
- Exactly half the values in a normally distributed dataset lie to the left of the mean, and the remaining half lie to the right.
- The Normal distribution is completely characterized by just two values: the mean and the standard deviation.
- The normal distribution curve is unimodal, meaning it has exactly one peak at the mean.

How to Implement in Excel

1. Input your complete dataset into an Excel spreadsheet.
2. Compute the mean of the dataset: `=AVERAGE(cell range)`
3. Compute the standard deviation of the dataset: `=STDEV(cell range)`
4. Select the value(s) for which you wish to compute the normal probability.
5. Apply the NORM.DIST function: `=NORM.DIST(x, mean, standard_deviation, FALSE)`

Dataset

The following table illustrates the normal distribution applied to employee incentive data.

Distribution is parameterized with **mean = 15,000** and **standard deviation = 3,500**.

Employee	Incentive (Rs.)	Normal Distribution f(x)
EMP1	5,000	1.92E-06
EMP2	7,000	8.36E-06
EMP3	9,000	2.62E-05
EMP4	10,000	4.11E-05
EMP5	11,000	5.93E-05
EMP6	12,000	7.89E-05
EMP7	13,000	9.68E-05
EMP8	14,000	1.09E-04
EMP9	15,000	1.14E-04
EMP10	16,000	1.09E-04
EMP11	17,000	9.68E-05
EMP12	18,000	7.89E-05

EMP13	19,000	5.93E-05
EMP14	21,000	2.62E-05
EMP15	23,000	8.36E-06

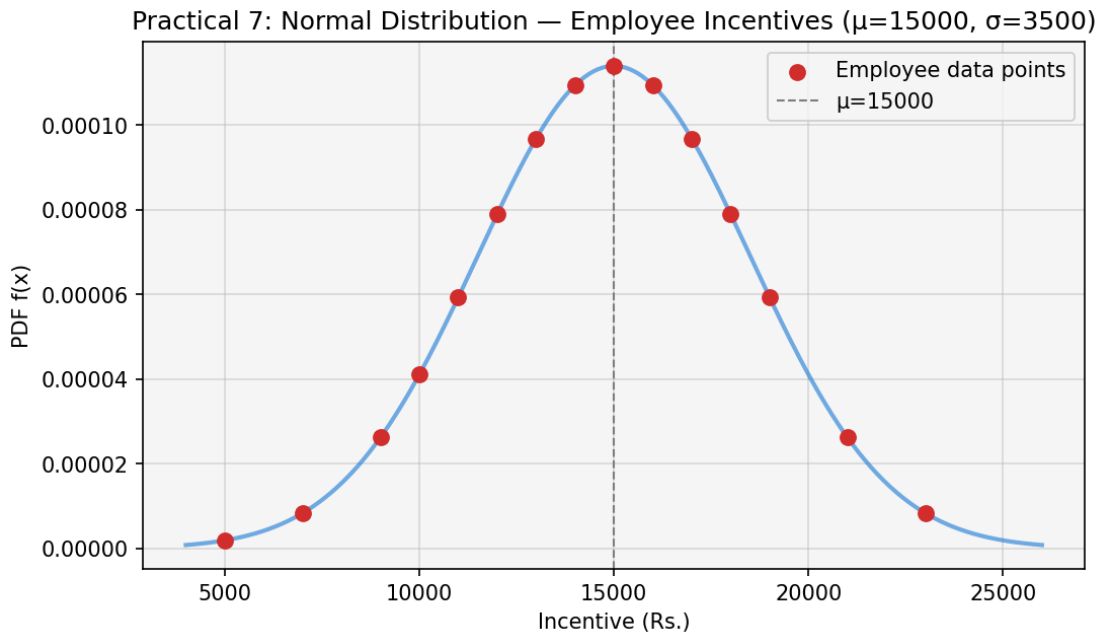

Figure 7: Normal Distribution PDF — Employee Incentives (μ=15000, σ=3500)

Excel formula: `=NORM.DIST(x, 15000, 3500, FALSE)`

The values peak at EMP9 (incentive = 15,000, the mean) and taper symmetrically in both directions, consistent with the bell-curve shape.

Practical 8: Calculation of Cumulative Distribution Functions for Exponential and Normal Distributions

Theory

The cumulative distribution function (CDF) gives the probability that a random variable takes a value less than or equal to a specified threshold. The Excel functions for CDF computation use the argument TRUE (cumulative = TRUE) instead of FALSE.

```
Normal CDF in Excel:      =NORM.DIST(x, mean, standard_deviation, TRUE)
Exponential CDF in Excel: =EXPON.DIST(x, lambda, TRUE)
```

Dataset — Normal Distribution CDF (mu = 6000, sigma = 2800)

Employee	Incentive (Rs.)	CDF Normal
EMP1	1,000	0.037073
EMP2	2,000	0.076564
EMP3	3,000	0.141988
EMP4	4,000	0.237525
EMP5	5,000	0.360492
EMP6	6,000	0.500000
EMP7	7,000	0.639508
EMP8	8,000	0.762475
EMP9	9,000	0.858012
EMP10	10,000	0.923436

Excel formula: `=NORM.DIST(x, 6000, 2800, TRUE)`

Dataset — Exponential Distribution CDF (lambda = 0.0025)

Days	Exponential CDF (lambda=0.0025)
150	0.312711
200	0.393469
250	0.464739
300	0.527633
350	0.583138
400	0.632121
450	0.675348
500	0.713495
550	0.747160
600	0.776870

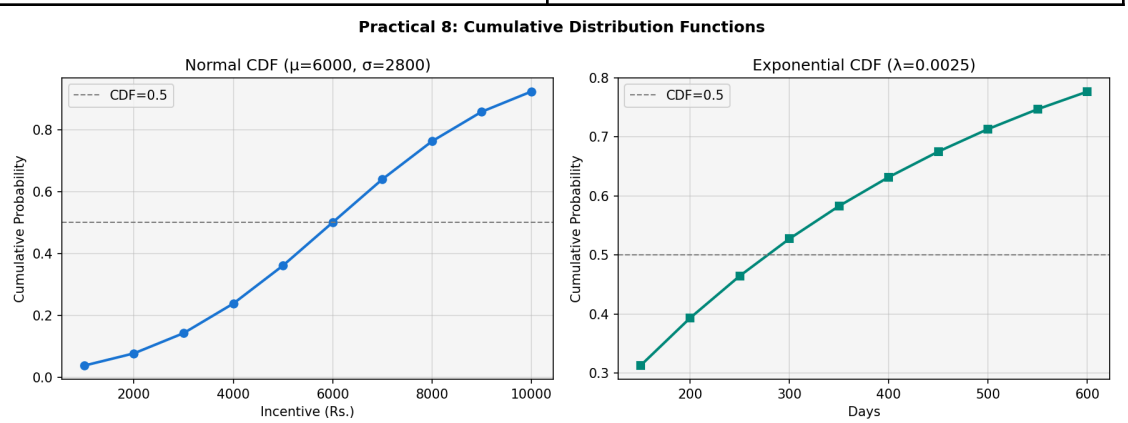

Figure 8: CDF Comparison — Normal (μ=6000, σ=2800) & Exponential (λ=0.0025)

Excel formula: `=EXPON.DIST(x, 0.0025, TRUE)`

The CDF values for both distributions are monotonically non-decreasing, approaching 1.0 as x tends to infinity.

Practical 9: Given Data from Two Distributions, Find the Distance Between the Distributions

Theory

Euclidean Distance

The Euclidean distance measures the straight-line separation between two points (or vectors) in a real-valued space. It is computed as the square root of the sum of the squared element-wise differences between two vectors.

```
Euclidean Distance = |X - Y| = sqrt[ Sum(i=1 to n) (xi - yi)^2 ]
```

Where:

- X: Array or vector X
- Y: Array or vector Y
- xi: Values along the horizontal axis in the coordinate plane
- yi: Values along the vertical axis in the coordinate plane
- n: Number of observations

How to Implement in Excel

Excel Formula: `=SQRT(SUMXMY2(array_X, array_Y))`

Dataset

The following table computes the Euclidean distance between the **Binomial distribution (n=8, p=0.6)** and the **Poisson distribution (lambda=4.8)** across 8 trial values:

Trials (k)	Binomial Dist. P(X=k)	Poisson Dist. P(X=k)	Squared Diff
1	0.007864	0.039503	0.00100099
2	0.041288	0.094807	0.00286428
3	0.123863	0.151691	0.00077438
4	0.232243	0.182029	0.00252148
5	0.278692	0.174748	0.01080439
6	0.209019	0.139798	0.00479151

7	0.089580	0.095862	0.00003946
8	0.016796	0.057517	0.00165818

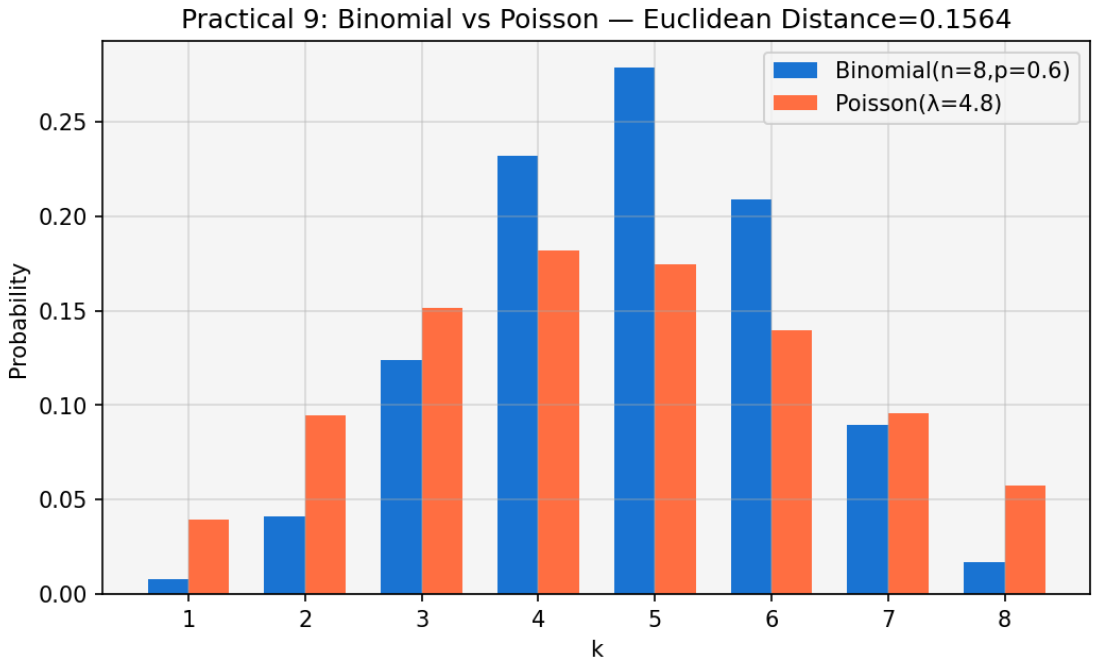

Figure 9: Euclidean Distance — Binomial(n=8,p=0.6) vs Poisson(λ=4.8)

Total Sum of Squared Differences: 0.02445467

Euclidean Distance = sqrt(0.02445467) = 0.156380

Excel formula: `=SQRT(SUMXMY2(B2:B9, C2:C9))`

As the number of trials increases toward k=4-5, both distributions show significant probability mass, then diverge most strongly at k=5, contributing the largest squared difference.

Practical 10: Application Problems Based on the Binomial Distribution

Theory

(Theory is identical to Practical 1 — Binomial Distribution. Refer to the formula, parameters, and Excel implementation described in Practical 1.)

Case Study — NEW: Smartphone Screen Defect Quality Control

Scenario

A smartphone manufacturing company claims that at most **8%** of its devices have a screen defect during production. A quality control team samples **6 devices** from an assembly line and finds **2 devices with screen defects**. The team wishes to evaluate whether this observation challenges the manufacturer's claim.

Parameters

- **Distribution:** Binomial
- **n = 6** (devices sampled)
- **p = 0.08** (claimed defect probability)
- **Observed:** k = 2 defective screens

Computation

Under the manufacturer's claim (p = 0.08), the probability of each outcome is:

k (Defective Screens)	**Binomial Distribution P(X=k)**
0	0.606355
1	0.316359
2 (Observed)	0.068774
3	0.007974
4	0.000520
5	0.000018
6	0.000000

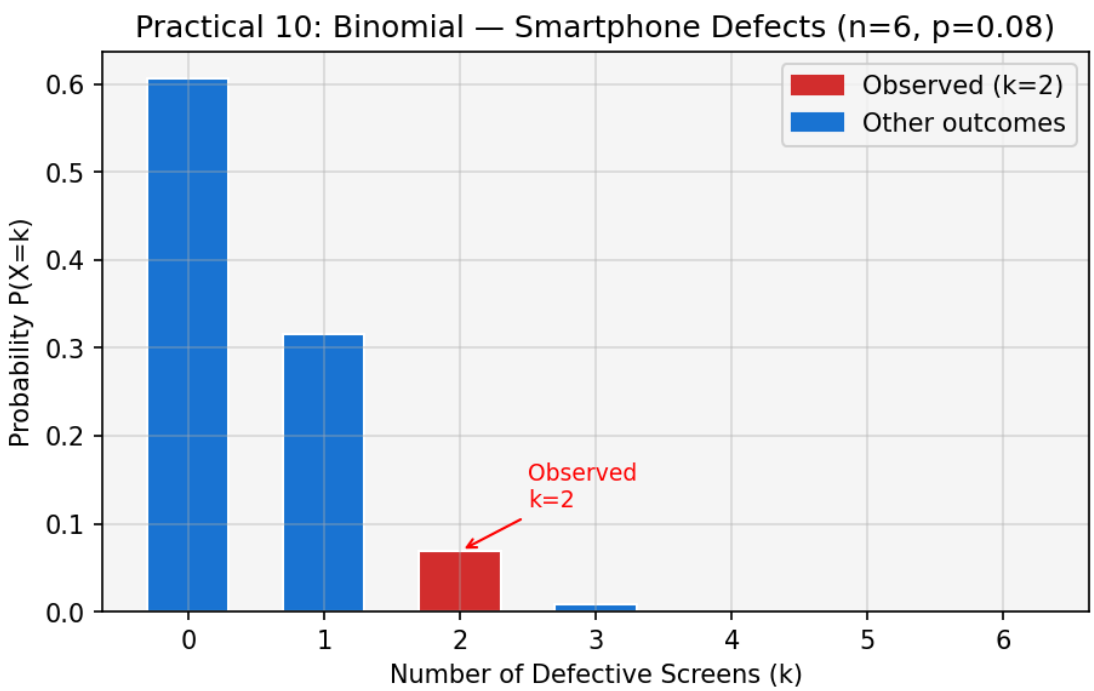

Figure 10: Binomial Application — Smartphone Defects (n=6, p=0.08)

Excel formula: `=BINOM.DIST(k, 6, 0.08, FALSE)`

P(X = 2) = C(6,2) x (0.08)^2 x (0.92)^4 = **0.068774** (approximately 6.88%)

Interpretation

Under the manufacturer's stated defect rate of 8%, finding exactly 2 defective screens out of 6 has a probability of approximately **6.88%**. While not impossible, this probability is relatively low. Furthermore, the cumulative probability P(X >= 2) = 1 - P(X=0) - P(X=1) = 1 - 0.606355 - 0.316359 = **0.077286** (about 7.7%). This suggests that observing 2 or more defects in a sample of 6, given p=0.08, is unlikely and may provide preliminary evidence that the true defect rate exceeds the manufacturer's claim of 8%.

Practical 11: Application Problems Based on the Poisson Distribution

Theory

(Theory is identical to Practical 3 — Poisson Distribution. Refer to the formula, parameters, and Excel implementation described in Practical 3.)

Case Study — NEW: Hospital Emergency Room Trauma Arrivals

Scenario

At a major urban hospital, the emergency room receives trauma cases at an average rate of **3.5 trauma arrivals per 2-hour shift**. The emergency coordinator wants to know the probability that **exactly 4 trauma cases** arrive during a single 2-hour shift to plan staffing resources.

Parameters

- **Distribution:** Poisson
- **lambda = 3.5** (average trauma arrivals per 2-hour shift)
- **Target:** Compute P(X = 4)

Computation

The Poisson probability table for x = 0 to 8:

x (Trauma Arrivals)	Poisson Distribution P(X=x)
0	0.030197
1	0.105691
2	0.184959
3	0.215785
4 (Answer)	0.188812
5	0.132169
6	0.077098
7	0.038549
8	0.016865

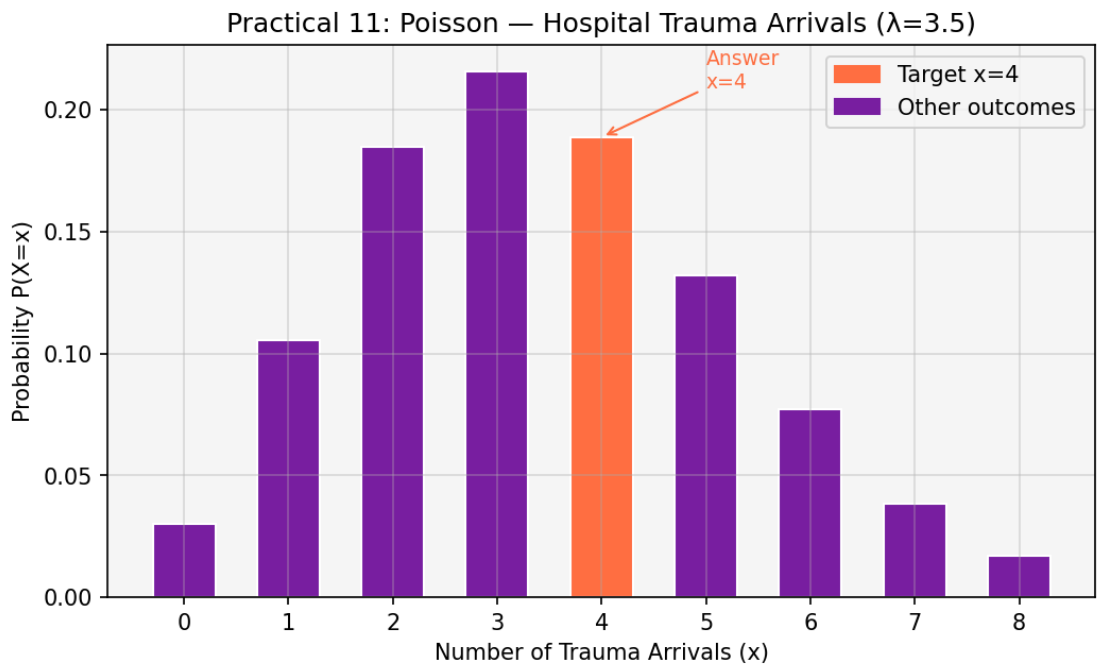

Figure 11: Poisson Application — Hospital Trauma Arrivals (λ=3.5)

Excel formula: `=POISSON.DIST(4, 3.5, FALSE)`

P(X = 4) = e^(-3.5) x 3.5^4 / 4! = **0.188812** (approximately 18.88%)

Interpretation

Given an average arrival rate of 3.5 trauma cases per 2-hour shift, the probability of receiving exactly 4 trauma arrivals in a shift is approximately **18.9%**. This is the second-highest individual probability (after x=3 at 21.6%), suggesting that shifts with 3 to 5 trauma arrivals are the most common scenarios. Hospital administrators should staff for at least 4-5 concurrent trauma cases to be well-prepared for the most probable demand levels.

Practical 12: Application Problems Based on the Normal Distribution

Theory

(Theory is identical to Practical 7 — Normal Distribution. Refer to the formula, parameters, and Excel implementation described in Practical 7.)

Case Study — NEW: Standardized Test Score Analysis

Scenario

A national educational assessment board conducts a standardized exam for university admissions. Historical data show that scores for a large cohort of students are approximately **normally distributed** with a **mean of 520** and a **standard deviation of 85**. An analyst wishes to compute the probability density at several score points to understand which scores are most probable and how scores are distributed across the cohort.

Parameters

- **Distribution:** Normal
- **mu = 520** (mean score)
- **sigma = 85** (standard deviation)

Computation

The table below shows seven selected scores and their corresponding normal PDF values:

Score (x)	Normal Distribution f(x)
400	0.00173260
450	0.00334366
500	0.00456530
520 (Peak — Mean)	0.00469344
540	0.00456530
580	0.00365842
640	0.00173260

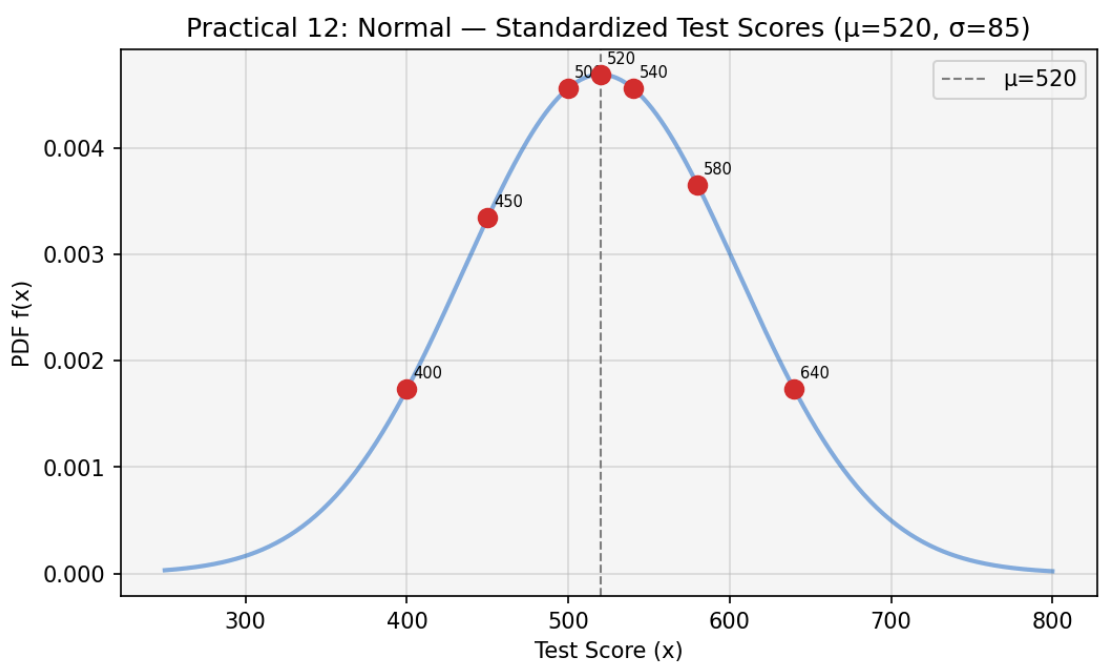

Figure 12: Normal Application — Standardised Test Scores (μ=520, σ=85)

Excel formula: `=NORM.DIST(x, 520, 85, FALSE)`

Interpretation

The PDF reaches its **maximum value of 0.00469344 at x = 520**, which is the mean. Scores of 500 and 540 (equidistant from the mean) yield identical densities (0.00456530), confirming the perfect symmetry of the normal curve. Scores at 400 and 640 — each 120 units from the mean (approximately 1.41 standard deviations) — also show equal densities of 0.00173260. These results confirm that the Normal distribution is bell-shaped and symmetric about its mean, with probability density decreasing smoothly as scores move away from 520 in either direction.

Practical 13: Covariance and Correlation — Concept and Computation

Theory

Covariance and correlation are two closely related statistical measures used to quantify the relationship between two random variables. While both describe the direction of a linear relationship, they differ in interpretation and scale.

Covariance Formula

```
Cov(X, Y) = Sum[(Xi - X_bar)(Yi - Y_bar)] / n
```

Where X_bar and Y_bar represent the sample means of the two datasets, and n is the number of paired observations.

Key Conceptual Distinctions between Covariance and Correlation

- Both datasets must contain the same number of observations.
- Neither dataset should be empty, and the standard deviation must not be zero.
- Covariance quantifies the extent to which two random variables tend to move in tandem.
- Correlation expresses how strongly two random variables are linearly associated with each other.
- Covariance is essentially an unscaled measure of correlation.
- Correlation is the standardized (scaled) form of covariance.
- Covariance can range from -infinity to +infinity, while correlation is bounded between -1 and +1.
- Covariance identifies only the direction of the linear relationship between variables.

- Correlation captures both the strength and direction of the linear relationship between two variables.
- Covariance is sensitive to changes in scale, whereas correlation is scale-invariant.

Pearson Correlation Coefficient Formula

```
r = Sum[(xi - x_bar)(yi - y_bar)] / sqrt[Sum(xi - x_bar)^2 * Sum(yi -
y_bar)^2]
```

Where:

- r = Pearson correlation coefficient
- xi = individual values of the x-variable in the sample
- x_bar = mean of the x-variable values
- yi = individual values of the y-variable in the sample
- y_bar = mean of the y-variable values

Types of Correlation

- **Positive Correlation:** As one variable increases, the other also increases.
- **Negative Correlation:** As one variable increases, the other decreases.
- **No Correlation:** The two variables exhibit no discernible linear relationship.

How to Implement in Excel

```
Pearson Correlation:   =PEARSON(array1, array2)  or  =CORREL(array1,
array2)
Population Covariance: =COVARIANCE.P(array1, array2)
```

Scatter Plots

A scatter plot is a two-dimensional graphical tool that displays the relationship between two numerical variables. Each observation is plotted as a point, with the independent variable on the x-axis and the dependent variable on the y-axis. Scatter plots are also called scatter graphs or scatter diagrams.

Scatter plots are especially useful in the following situations:

- When analyzing large datasets with many paired observations.
- When each data point is represented as a pair of numerical values.
- When both variables are measured on a continuous numerical scale.

Dataset — Study Hours vs. Test Score (7 paired values)

Study Hours (X)	Test Score (Y)
2	45
4	58
5	62
7	71
8	74

10	82
12	89

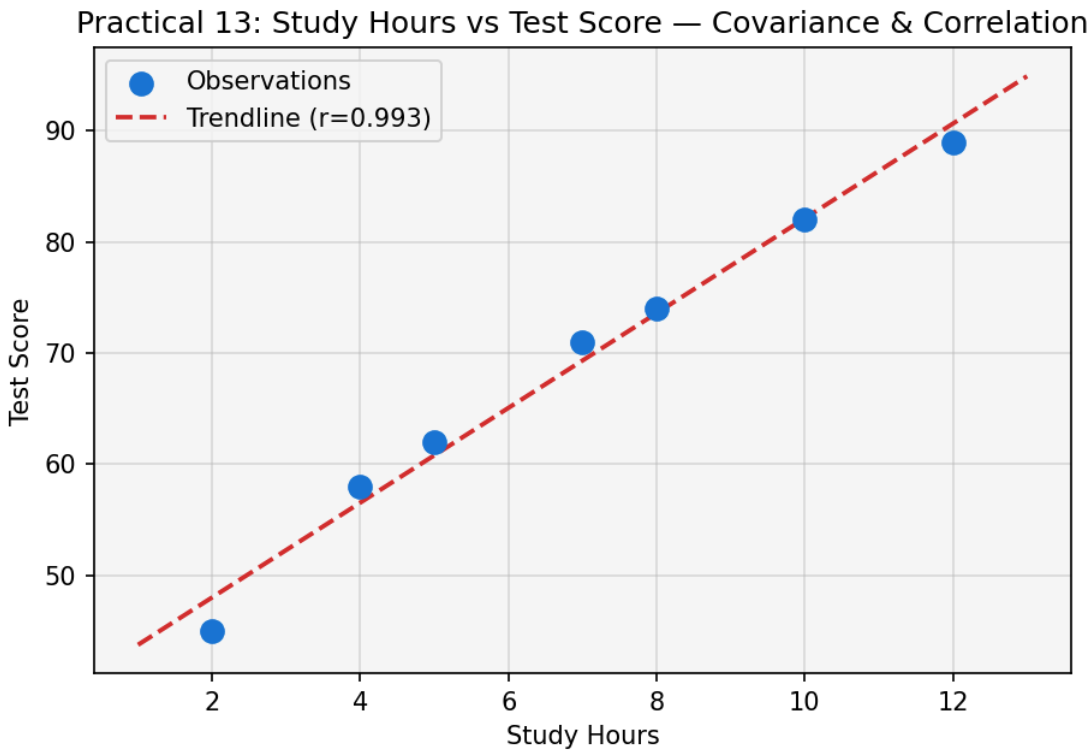

Figure 13: Scatter Plot — Study Hours vs Test Score (r=0.993)

Excel formula for Covariance: `=COVARIANCE.P(A2:A8, B2:B8)`

Excel formula for Correlation: `=CORREL(A2:A8, B2:B8)`

Summary Statistics:

- Mean Study Hours: 6.857
- Mean Test Score: 68.714
- Population Covariance: 44.39
- Pearson Correlation (r): 0.993

The covariance of approximately 44.39 indicates a strong positive relationship between study hours and test scores.

To insert a scatter chart in Excel: navigate to the **Insert** tab > **Charts** group > **Scatter Chart**.

Practical 14: Calculation of Karl Pearson's Correlation Coefficients

Theory

Karl Pearson's correlation coefficient is a standard measure of the linear association between two variables. It takes values in the range [-1, +1], where -1 indicates a perfect negative linear relationship, 0 indicates no linear association, and +1 indicates a perfect positive linear relationship.

(Refer to the Pearson Correlation Coefficient Formula and Excel implementation in Practical 13.)

Dataset

The following dataset presents the values of x alongside three corresponding y-variables (Y1, Y2, Y3):

x	Y1	Y2	Y3
3	85	20	55
5	72	32	42
7	60	45	68
9	48	58	44
11	35	70	51

Excel formula: `=CORREL(A2:A6, B2:B6)` *for XY1, etc.*

Correlation Results

Pair	Correlation Coefficient r	Type
X vs Y1	-0.9999	Strong Negative Correlation
X vs Y2	+0.9999	Strong Positive Correlation
X vs Y3	-0.0915	Near-Zero / No Correlation

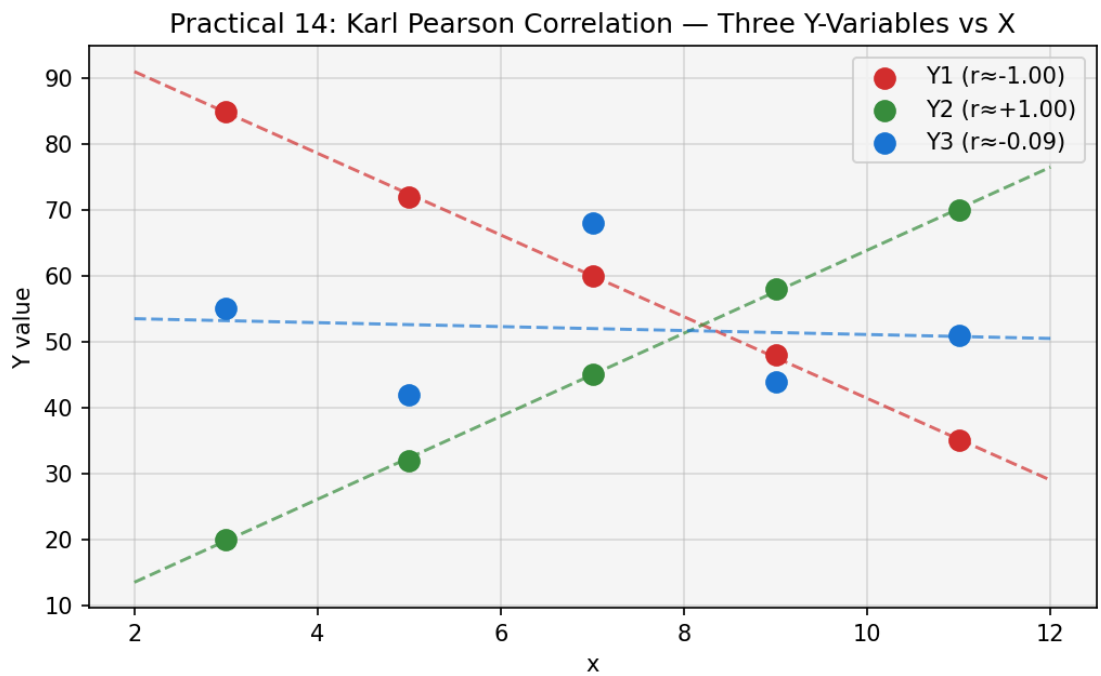

Figure 14: Karl Pearson Correlation — Three Y-Variables vs X

Interpretation: XY1 shows a near-perfect negative correlation (-1.00), XY2 shows a near-perfect positive correlation (+1.00), and XY3 shows virtually no linear correlation (-0.09).

Practical 15: Finding the Correlation Coefficient for a Bivariate Frequency Distribution

Theory

A bivariate frequency distribution tabulates the joint frequency of two variables simultaneously. It enables the computation of the correlation coefficient directly from the frequency table, providing insight into the linear relationship between the two grouped variables.

Dataset — New 5x4 Contingency Table

The following table presents a bivariate frequency distribution of student **Marks (rows)** against **Age in Years (columns)**:

Marks \ Age	**16-18**	**18-20**	**20-22**	**22-24**	**Total**
10-20	3	2	1	0	6
20-30	4	3	2	1	10
30-40	2	5	6	3	16
40-50	1	3	5	5	14
50-60	0	1	2	3	6
Total	10	14	16	12	52

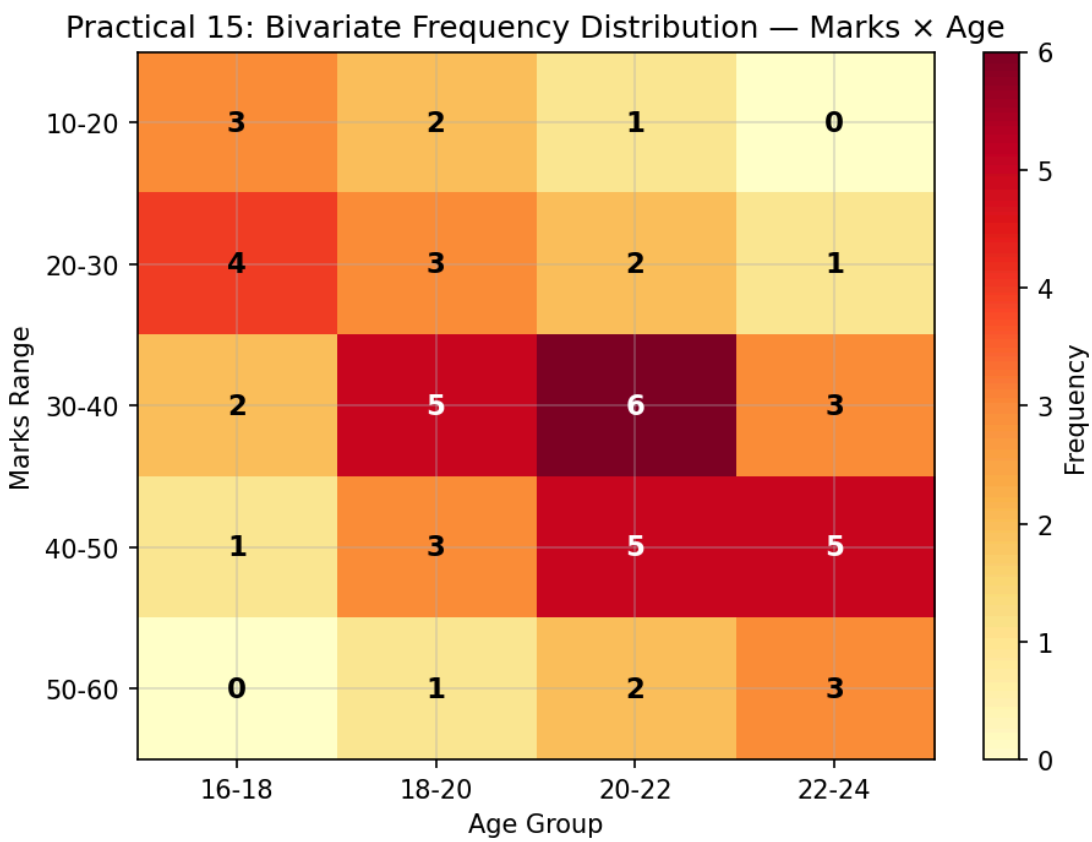

Figure 15: Bivariate Frequency Heatmap — Marks × Age Group

Computed Results:

- Correlation coefficient along the x-axis (Age): -0.187
- Correlation coefficient along the y-axis (Marks): +0.775

The positive and relatively strong y-axis correlation (0.775) indicates that marks tend to increase with age in years, while the x-axis shows a weak negative correlation.

Practical 16: Generating Random Numbers from Discrete Distributions (Bernoulli, Binomial, Poisson)

Theory

Random number generation from discrete probability distributions is a core computational task in simulation and statistical modeling. Excel provides built-in functions for generating random values conforming to discrete distributions such as the Binomial.

How to Implement in Excel

For Binomial Distribution: `=BINOM.INV(1, P, RAND())`

This formula generates a random number equal to 1 with probability P, and 0 otherwise. To generate a random binomial count from n trials, use: `=BINOM.INV(n, P, RAND())`

Dataset — Example Random Numbers

Example random outputs from **Binomial distribution with n = 10 and p = 0.35:**

Using formula: `=BINOM.INV(10, 0.35, RAND())`

Run	Random Number from B(10, 0.35)
1	3
2	5
3	2
4	4
5	3

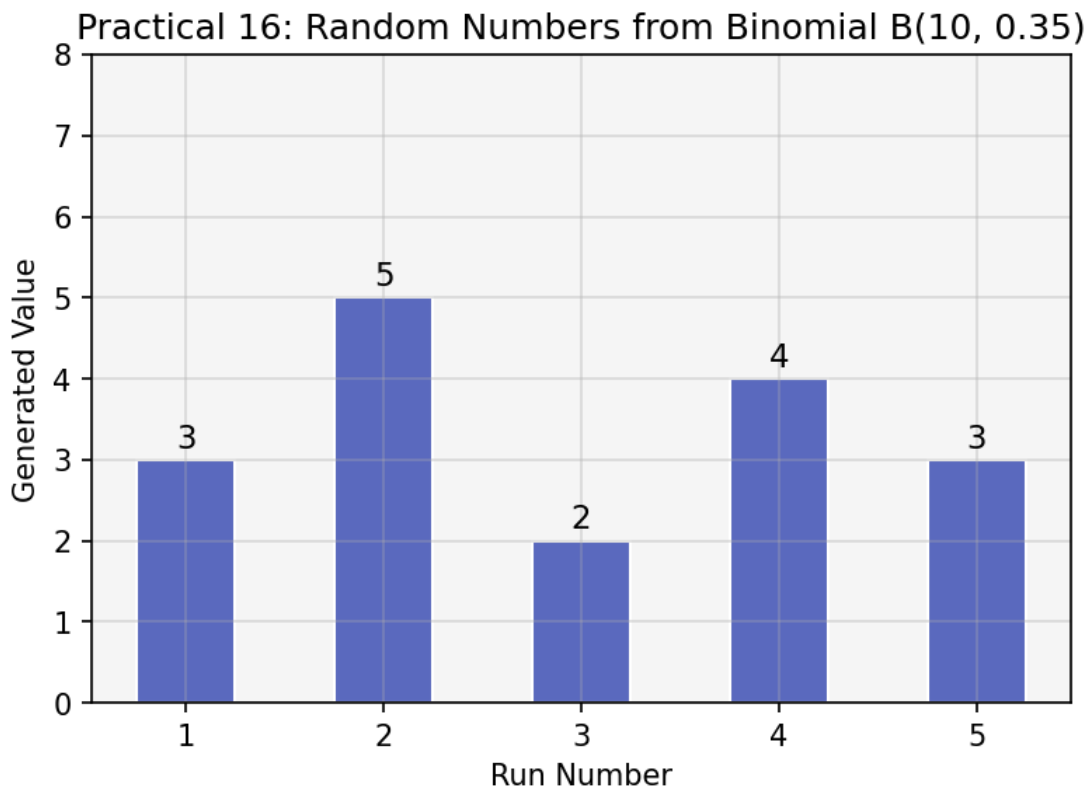

Figure 16: Random Numbers from Binomial Distribution B(10, 0.35)

Each execution of the formula produces a different random binomial value. These values approximate the distribution B(10, 0.35).

Practical 17: Generating Random Numbers from Continuous Distributions (Uniform, Normal)

Theory

Continuous distributions such as the Uniform and Normal distributions require different approaches for random number generation. Excel supports this through the NORMINV function combined with the RAND() function.

How to Implement in Excel

For Normal Distribution: `=NORMINV(RAND(), mean, standard_deviation)`

RAND() generates a uniform random probability between 0 and 1. NORMINV then maps this probability to the corresponding value from the specified Normal distribution.

For Uniform Distribution: `=a + RAND() * (b - a)`

This generates a uniformly distributed random number between a and b.

Dataset — Example Random Numbers

Example random outputs from **Normal distribution with mean = 25 and SD = 6:**

Using formula: `=NORMINV(RAND(), 25, 6)`

Run	Random Number from N(25, 6^2)
1	23.7152
2	28.4391
3	20.1867
4	30.2543
5	24.8619

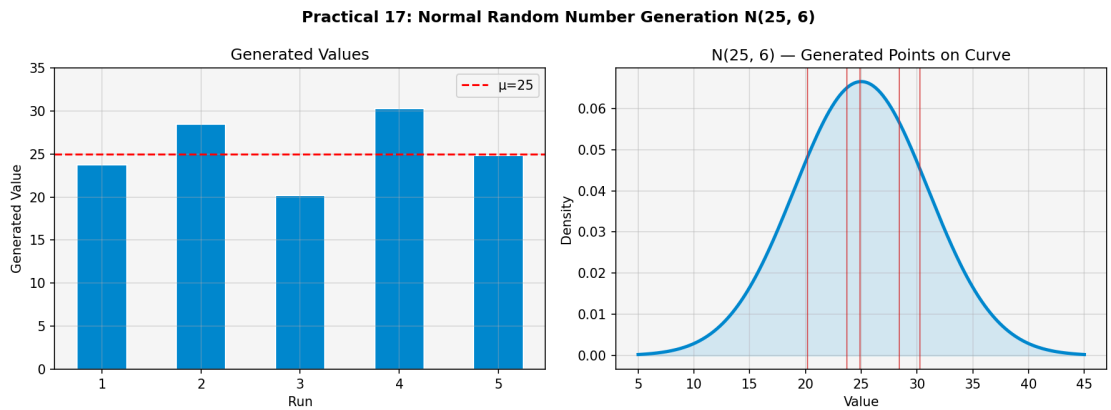

Figure 17: Random Numbers from Normal Distribution N(25, 6)

These randomly generated values approximate a Normal distribution N(25, 6^2) and will differ with each worksheet recalculation.

Practical 18: Finding the Entropy from a Given Dataset

Theory

The entropy of a random variable represents the average degree of information, surprise, or uncertainty embedded in the variable's possible outcomes. For a discrete random variable X that assumes values in the alphabet x and is distributed according to the probability mass function P(x) in [0, 1], the entropy H(X) is defined as:

```
H(X) = -Sum p(x) * log2[p(x)]     for x in X
```

Important Notes on the Choice of Logarithm Base

- **Base 2:** yields entropy in units of **bits**.
- **Base e (natural logarithm):** yields entropy in natural units (**nats**).
- **Base 10:** yields entropy in **hartleys** (also known as bans or dits).

An equivalent interpretation of entropy is that it equals the expected value of the self-information of a random variable.

Dataset — New: Four-Outcome Information Source

The following table computes entropy for a discrete random variable X with 4 possible outcomes (A, B, C, D):

Outcome	**P(x)**	**log2 P(x)**	**-P(x) * log2 P(x) = H(x)**
A	0.40	-1.3219	0.5288
B	0.30	-1.7370	0.5211
C	0.20	-2.3219	0.4644
D	0.10	-3.3219	0.3322

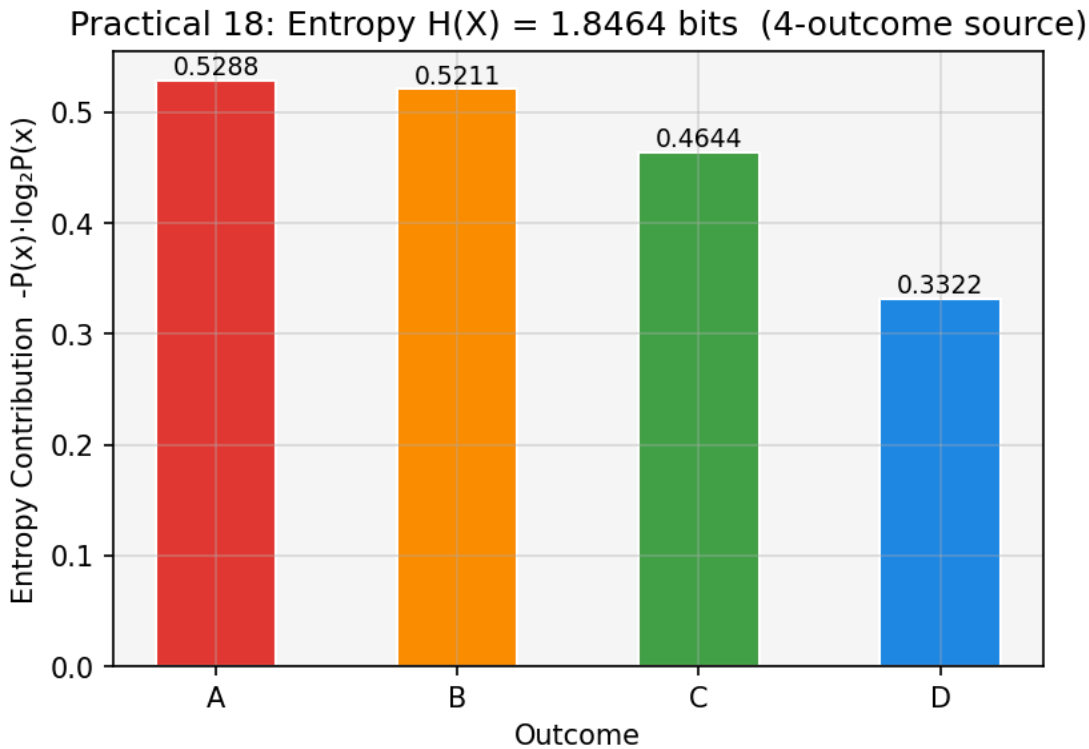

Figure 18: Entropy Contributions by Outcome — H(X)=1.8464 bits

Sum of Probabilities: 0.40 + 0.30 + 0.20 + 0.10 = **1.00** (checksum verified)

Total Entropy H(X) = 0.5288 + 0.5211 + 0.4644 + 0.3322 = 1.8464 bits

> ***Excel formula (Outcome A):*** `=-0.4 * LOG(0.4, 2)` *or* `=-0.4 * LN(0.4) / LN(2)`
>
> ***Total Entropy:*** `=SUMPRODUCT(-prob_range, LOG(prob_range, 2))`

Interpretation

The computed entropy H(X) = **1.8464 bits** indicates a moderate level of uncertainty in this 4-outcome system. For reference:

- Maximum possible entropy for 4 equally likely outcomes = log2(4) = **2.0 bits** (when all P = 0.25).
- The unequal probabilities (0.40, 0.30, 0.20, 0.10) reduce entropy below the maximum, as outcome A is predictably more likely.
- The system is more uncertain than a deterministic outcome (0 bits) but less uncertain than a uniform 4-outcome variable.